Progress Report

Multi-Dimensional Preference Annotation of Advertising Interviews: LLM-Based Coding with Mistral-7B

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Abstract—This report presents the design and execution of a multi-dimensional preference annotation pipeline applied to a corpus of 800 qualitative advertising interviews. Each panelist responded to six comparative questions about two telecom advertisements: Bouygues Telecom (absurd humour, crime-scene parody) and Orange (network reliability, reassuring tone). Using a locally deployed Mistral-7B-Instruct-v0.2 model via Ollama on Apple MPS, each interview is annotated across three dimensions — *creative preference*, *commercial preference* and *overall preference* — using a structured few-shot prompting strategy with a constrained JSON output schema. The full run completes with zero parse errors and a mean annotation confidence of 0.966. Results reveal a strong creative/commercial dissociation: Bouygues dominates creative preference (60%) while Orange dominates commercial (83.8%) and overall (71%) preference.

1. OBJECTIVE

Previous stages of this project established lexical distance metrics and brand-specific vocabulary distributions. This stage introduces **structured preference annotation**: each panelist's full set of responses is assigned categorical labels along three theoretically motivated dimensions. The goal is to quantify, at scale, whether the two campaigns succeed differently across creative appeal, commercial persuasiveness and overall brand impact.

2. ANNOTATION SCHEMA

Three dimensions are defined, each taking values from {*Bouygues*, *Orange*, *Neutral*, *Mixed*}:

- **creative_preference** — preferred brand for creativity, humour, originality and storytelling.
- **commercial_preference** — preferred brand for reliability, trust and purchase intent.
- **overall_preference** — global preferred brand taking all dimensions into account.

Mixed indicates balanced appreciation; *Neutral* indicates no expressed opinion. Each annotation also includes a `reasoning` field and a scalar `confidence` score in $[0, 1]$.

3. MODEL AND PROMPTING

Annotations are produced by **Mistral-7B-Instruct-v0.2** deployed locally via Ollama on Apple MPS, averaging 6–10 seconds per inference. A HuggingFace-loaded version was initially attempted but abandoned after a MPS out-of-memory error (`MPS allocated: 18.03 GiB`, exceeding the available pool). Ollama resolves this by managing memory independently of the PyTorch MPS allocator.

The system prompt defines the annotator role and full schema. Four few-shot gold examples are prepended to each user prompt, balanced to cover all valid output configurations:

1. Bouygues creative, Orange commercial, Mixed overall
2. Bouygues on all three dimensions
3. Orange on all three dimensions
4. Bouygues creative, Mixed commercial, Bouygues overall

An earlier three-example set (containing Orange in all commercial slots) produced label collapse with Orange dominating nearly all commercial annotations. Adding examples 2 and 4 resolved this bias, confirmed by the pilot distribution.

Continuation of the lexical structure and adjective-distance analyses. This report documents the design, implementation and results of an automated multi-dimensional preference annotation pipeline applied to the full 800-panelist interview corpus using a locally deployed large language model.

4. PILOT VALIDATION

A 30-sample pilot was executed prior to the full run. Table 1 summarises the pilot label distribution following few-shot rebalancing.

Table 1. Pilot distribution (n=30).

Dimension	Bouygues	Orange	Other
Creative	19 (63%)	11 (37%)	—
Commercial	1 (3%)	26 (87%)	3 (10%)
Overall	5 (17%)	25 (83%)	—

Orange's commercial dominance in the pilot was validated against the known corpus structure: panelists consistently invoke *fiabilité* and reassurance as Orange's key attributes, making this a substantive rather than a model-induced signal. An expert spot-check of 10 verbatims confirmed alignment between model outputs and qualitative reading in ≥8/10 cases.

5. FULL ANNOTATION RUN

The full run processes all 800 panelists with a checkpoint every 50 samples. Table 2 summarises performance.

Table 2. Full run statistics (n=800).

Metric	Value
Total samples	800
ParseErrors	0 (0.0%)
Uncertain labels	0 (0.0%)
Low confidence (<0.7)	0
Mean confidence	0.966
Avg. inference time	≈8.5–10.6 s/sample
Total wall-clock time	≈80 min

Figure 1 shows that confidence values cluster tightly at 0.95 and 0.98, corresponding to unambiguous and moderately ambiguous verbatims respectively. Zero parse errors across 800 samples confirms the robustness of the constrained JSON schema.

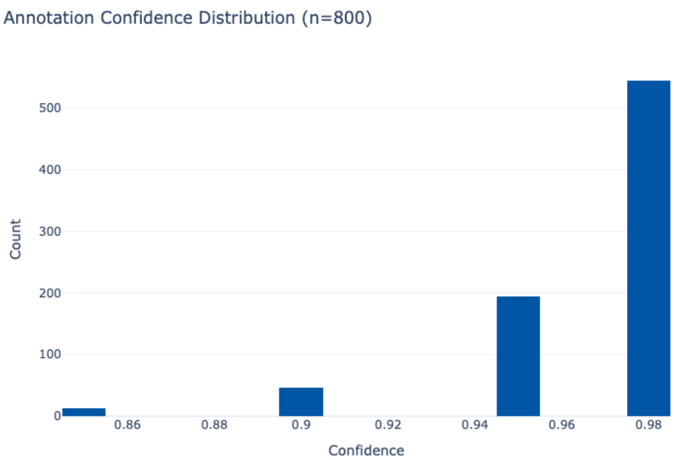

Figure 1. Distribution of annotation confidence scores (n=800). Values are tightly concentrated at 0.95 and 0.98, with a mean of 0.966.

6. RESULTS

6.1. Preference Distributions

Table 3 presents the full label distribution across all three dimensions.

Table 3. Label distribution across annotation dimensions (n=800).

Dimension	Bouygues	Orange	Mixed	Neutral
Creative	60.0%	34.0%	6.0%	0.0%
Commercial	4.6%	83.8%	10.9%	0.8%
Overall	19.4%	71.0%	9.6%	0.0%

Figure 2 visualises these distributions as a stacked bar, making the reversal between creative and commercial preference immediately apparent.

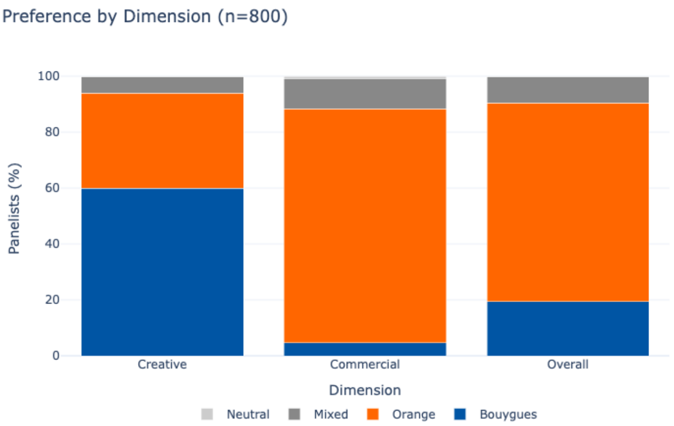

Figure 2. Stacked preference distribution across three annotation dimensions (n=800). Bouygues dominates creative preference (60%), while Orange dominates commercial (83.8%) and overall (71%).

6.2. The Creative–Commercial Dissociation

The central finding is a strong dissociation between creative and commercial preference. Bouygues wins creatively by a wide margin (60% vs. 34%), while Orange dominates both commercial (83.8% vs. 4.6%) and overall (71% vs. 19.4%) dimensions.

Figure 3 quantifies this dissociation at the individual level. Among the 60% of panelists who prefer Bouygues creatively, only 32% select Bouygues as their overall preference — while 52% still choose Orange overall despite their creative admiration for the competing brand. Panelists who prefer Orange creatively convert to Orange overall at 100%.

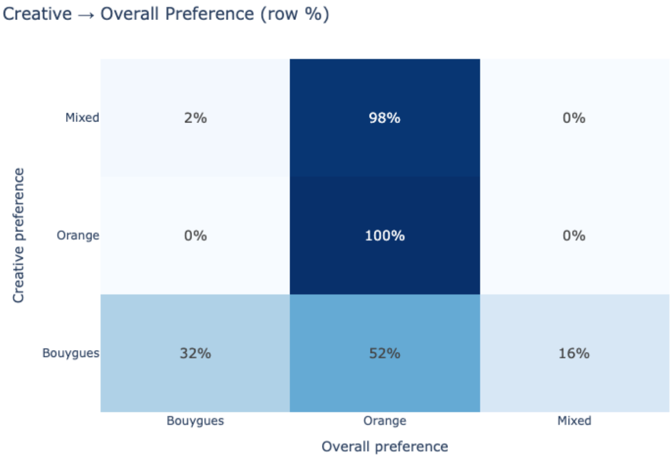

Figure 3. Row-normalised transition matrix from creative to overall preference. Among Bouygues creative fans, 52% still select Orange overall — illustrating the failure of creative impact to convert into commercial preference.

The 10.9% *Mixed* commercial label identifies a non-trivial undecided segment — panelists who found both campaigns commercially credible — representing a potential conversion opportunity for Bouygues.

6.3. Overall Preference: Aggregate View

Figure 4 shows the overall preference share as a proportional summary. Orange captures 71% of overall preference, with Bouygues at 19.4% and a mixed segment of 9.6%.

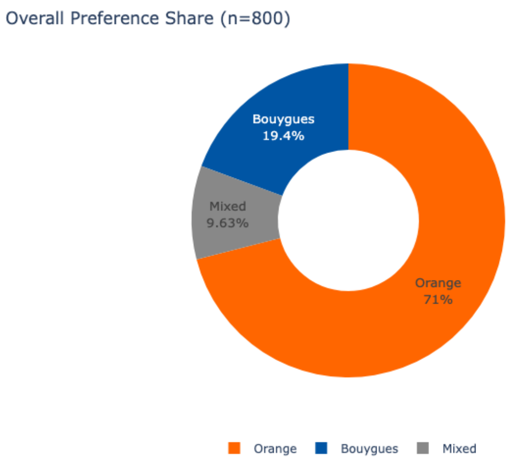

Figure 4. Overall preference share across 800 panelists. Orange captures nearly three quarters of overall preference despite Bouygues winning the creative dimension.

7. DEMOGRAPHIC ANALYSIS

7.1. Age

Figure 5 reveals a strong age gradient in overall preference. Panelists aged 18–39 show substantial Bouygues overall preference and high *Mixed* rates, reflecting genuine ambivalence between the two campaigns. From age 40 onwards, Orange dominance intensifies sharply, reaching approximately 90% in the 50–65 bracket. This suggests that Orange's reliability messaging resonates most strongly with older segments, while Bouygues retains relative traction among younger audiences.

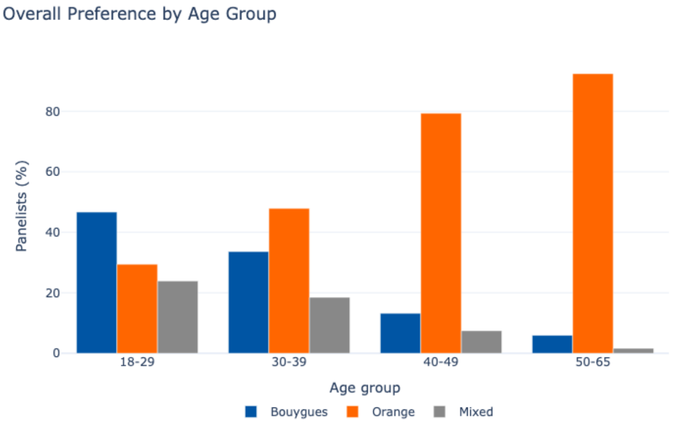

Figure 5. Overall preference by age group. Orange dominance intensifies with age; Bouygues retains the strongest relative share in the 18–29 cohort.

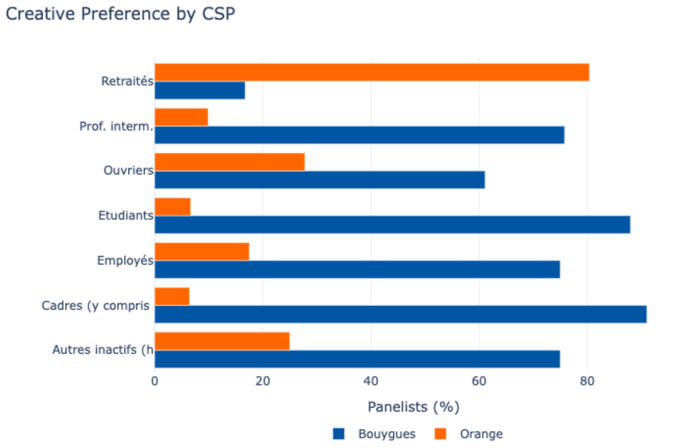

Figure 7. Creative preference by socio-professional category (CSP). Bouygues wins creatively in all CSP groups except *Retraités*, where Orange achieves its strongest creative showing.

7.2. Gender

Figure 6 shows the rate at which each gender group exhibits a creative–overall split (i.e., prefers Bouygues creatively but a different brand overall). All three groups show split rates around 45–58%, confirming that the creative/commercial dissociation is not gender-specific but a corpus-wide phenomenon. Non-binary respondents show a marginally higher split rate, though the subgroup is small.

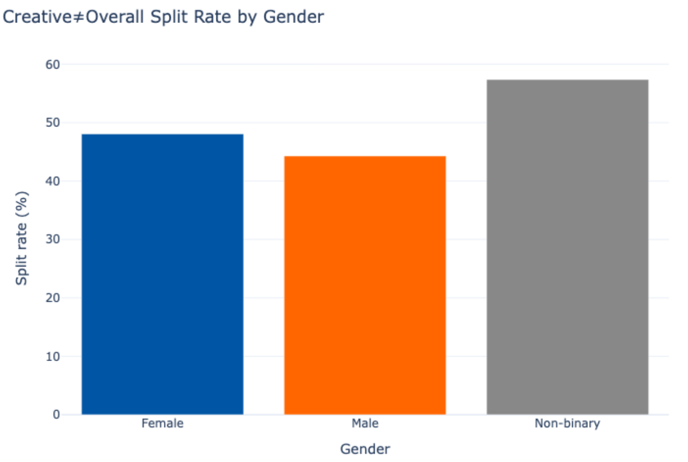

Figure 6. Percentage of panelists per gender whose creative preference differs from their overall preference. The split rate is consistently high across all groups (>44%), confirming the dissociation is gender-independent.

7.3. Socio-Professional Category

Figure 7 breaks down creative preference by socio-professional category (CSP). Bouygues creative preference is uniformly high across all CSP groups (60–90%), with *Cadres* and *Étudiants* showing the strongest creative alignment with Bouygues. Orange creative preference is notably higher among *Retraités* (approximately 80%), consistent with the age gradient observed in Figure 5 — retired panelists are likely to fall in the older cohorts where Orange's reassuring tone resonates more broadly.

8. INTERPRETATION

Taken together, the annotation results reveal a textbook **creative–commercial dissociation**:

- **Bouygues** achieved strong creative impact through its absurd humour and crime-scene parody format. Its originality was widely recognised across all demographic groups. However, this creative success did not translate into commercial persuasion or overall brand preference.
- **Orange** achieved the inverse: its reliability-centred message was less creatively distinctive but consistently converted into commercial preference and overall brand choice, particularly among older and less urban panelists.
- The **individual-level transition matrix** (Figure 3) is the most telling result: among Bouygues creative fans, a majority (52%) still selected Orange overall. Orange's commercial message overrode Bouygues' entertainment value at the moment of preference formation.

This pattern is consistent with established advertising theory distinguishing *likeability* from *persuasion*. A highly entertaining advertisement does not necessarily generate stronger purchase intent than a simpler, benefit-driven message — especially for a product category (mobile network) where reliability is a primary purchase driver.

9. OUTPUT FILES

The annotation pipeline produces the following artefacts:

- `preference\_annotations\_multidim.csv` — 800-sample annotation table with labels, reasoning, confidence and flag columns.
- `interviews\_with\_multidim\_preferences.csv` — original panel (800×48) merged with annotation columns (final shape: 800×54).
- `preference\_summary\_table.csv` — percentage distribution table across all dimensions.
- `annotations\_checkpoint.csv` — incremental checkpoint saved every 50 samples.
- `pilot\_annotations\_30.csv` and `pilot\_to\_annotate\_manually.csv` — pilot outputs for validation.

10. SUMMARY

This notebook implements a complete LLM-based annotation pipeline for multi-dimensional preference coding of 800 advertising interviews.

The pipeline demonstrates zero parse errors, high annotation confidence (mean: 0.966) and a substantively coherent output distribution validated against domain knowledge of the corpus. The central empirical result — Bouygues wins creatively, Orange wins commercially and overall — provides a structured quantitative foundation for subsequent cross-tabulation, demographic modelling and integration with the lexical distance metrics established in earlier stages of the project.